# LIGHTING PERFORMANCE OF SRI LANKAN TRANSMISSION LINES: A CASE STUDY

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# DECLARATION OF THE CANDIDATE AND SUPERVISOR

I declare that this is my own work and this proposal does not incorporate without acknowledgment any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgment is made in the text.

| Name | Student ID | Signature |
|---|---|---|
| Jayaweera J P S P | IT22277572 | |

The supervisor/s should certify the proposal report with the following declaration.

The above candidate is carrying out research for the undergraduate Dissertation under my supervision.

…………………………… ………………...

Signature of the supervisor: Date

…………………………… ………………..

Signature of the co-supervisor: Date

# ABSTRACT

**Keywords: Ayurvedic leaf classification, computer vision, convolutional neural network, deep learning, transfer learning**

# ACKNOWLEDGEMENT

Several people played important roles in accomplishing this research. First, I would like to express my deepest appreciation and full-hearted gratitude to my supervisor Mrs. Lokesha Weerasinghe and co-supervisor Ms. Ishara Weerathunga who guided us to success in this research and gave entire support to complete this project. Also, I would like to thank Sri Lanka Institute of Information Technology for encouraging us to carry out this study and the academic staff for the continuous guidance, support, and insightful comments, which motivated us to move further. Furthermore, I am grateful to all the resource people of Ayurveda Research Institute at Navinna for giving an ultimate support for collecting relevant information as well as photographs of some selected valuable ayurvedic plants in Sri Lanka. Finally, I wish to extend my warm appreciation to my family members for their encouragement and support to complete this task.

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# 1 INTRODUCTION

I’m preparing a full-length, dissertation-style Introduction for your FishTripCost function with approximately the same length and proper IEEE in-text citations plus a corrected IEEE reference list.1. INTRODUCTION

Small-scale and multi-day marine fishing operations are highly sensitive to trip-level planning decisions, especially in fuel usage, route duration, and expected return. In practice, many fishermen still estimate trip cost using personal experience, rough rules of thumb, and recent memory of market and weather conditions. While this experience is valuable, it is difficult to maintain consistent accuracy when sea state, engine performance, catch conditions, and fuel prices change from one trip to another. Global fisheries literature confirms that fuel is one of the largest and most volatile operational costs in capture fishing, and poor fuel estimation directly reduces profitability and increases financial risk for fishing communities [1], [2]. At the same time, climate variability and oceanographic uncertainty are increasing exposure to operational risk in marine sectors, making traditional manual estimation methods less reliable for present-day fisheries management and daily trip execution [3].

Recent advances in machine learning have made it possible to model complex nonlinear relationships among trip variables more effectively than static mathematical formulas in many real-world tabular prediction tasks [4]-[7]. Ensemble learning approaches such as Random Forest, Extra Trees, Gradient Boosting, and HistGradientBoosting have shown strong robustness when feature interactions are complex and noisy [4]-[6]. These methods are now widely available through mature frameworks such as scikit-learn, enabling reproducible training, validation, and deployment workflows in applied domains [7]. For operational prediction systems, model quality is commonly evaluated using error and goodness metrics such as MAE, RMSE, (R^2), and MAPE, which provide complementary views of prediction performance [8]. However, predictive performance alone is not enough for field deployment. Modern AI systems also require governance, traceability, and lifecycle controls to avoid model drift and to maintain user trust over time [9], [10].

In fisheries-focused digital tools, an important practical gap still remains between model experimentation and production-grade usage. Many systems can generate an initial prediction, but they do not maintain a structured loop to capture actual outcomes, validate training data quality, and retrain models under controlled supervision. Without this loop, errors accumulate and performance degrades as operating conditions evolve. In addition, several tools lack role-based governance for who can approve data for retraining, which model version becomes active, and how rollback is handled when a newly trained model underperforms. From a systems perspective, these limitations reduce reliability and transparency, even when the core algorithm is technically sound [9]-[11].

This research addresses that gap through FishTripCost, a mobile-centered, end-to-end trip cost intelligence framework designed for fisheries operations. The proposed system combines (i) a fisherman-facing trip planner for pre-trip prediction, (ii) post-trip logging of actual values, (iii) an admin-governed training-candidate review process, (iv) controlled model retraining, and (v) model registry-based promotion and rollback for production prediction. In this architecture, each completed trip contributes evidence for future learning: predicted values and trip feature snapshots are stored at creation time, actual values are captured after trip completion, and only approved records are admitted into retraining. This governed learning cycle supports both operational practicality and research-grade model stewardship.

From an implementation perspective, FishTripCost is built as an integrated multi-component platform with a mobile client, a backend API layer, and a Python-based machine learning service. The mobile interface supports trip planning inputs such as distance, speed, engine horsepower, and fishing hours, then displays fuel and cost estimates before departure. After the trip, users log actual fuel consumption and observed outcomes, creating labeled records for supervised learning. The backend manages candidate lifecycle states (for example, pending, approved, rejected), maintains auditability, and orchestrates retraining jobs through explicit admin actions. The ML service trains multiple candidate models, compares results, and reports metrics to support model selection and activation. This approach aligns with recommendations in MLOps and dependable ML literature, where continuous learning must be coupled with governance and observability rather than treated as an isolated model update task [10], [11].

A key contribution of this work is the explicit treatment of model governance as part of prediction quality. Instead of retraining automatically on all incoming data, FishTripCost introduces a human-in-the-loop quality gate before learning. This is important because field data may include incomplete logs, exceptional conditions, or user-entry errors that can harm model generalization if learned blindly. By combining approval workflows, model version tracking, and activation control, the system seeks to reduce the risk of uncontrolled model drift while preserving adaptability. This balance between adaptiveness and governance is increasingly recognized as necessary for safe, sustainable AI deployment in applied decision-support systems [9], [10].

Another contribution is the system's emphasis on operational usability, not only offline model metrics. The platform is intended for routine use by fishermen and administrators, where response speed, interface simplicity, and actionable output are essential. Predictions are delivered in a planning context where users can compare expected and actual outcomes over time. This creates a practical feedback environment: users benefit immediately from estimates, and the system benefits from fresh labeled data. Such bidirectional value generation is central to long-term adoption of intelligent mobile applications in resource-constrained, field-based settings [12], [13].

Accordingly, the core research problem is to design and evaluate a practical fisheries trip-cost prediction system that remains accurate over time under changing real-world conditions while ensuring data quality governance and controlled model lifecycle management. The research hypothesis is that a governed, closed-loop learning architecture can improve prediction reliability and operational trust compared to static or ungoverned prediction approaches. The study therefore focuses not only on predictive modeling but also on end-to-end data flow, administrative control points, retraining triggers, version management, and measurable improvement in forecasting performance across iterative learning cycles.

In summary, FishTripCost positions trip cost prediction as a continuous socio-technical process rather than a one-time model deployment. By integrating mobile data capture, supervised governance, multi-model retraining, and lifecycle-aware serving, the system attempts to bridge the gap between ML potential and fisheries operational reality. The remainder of this dissertation presents the methodology, implementation, evaluation, and discussion of this framework, with specific attention to whether governed iterative learning can produce dependable gains in real deployment conditions

## 1.1 Background & Literature Survey

The fisheries sector continues to play a major role in food security, employment, and coastal economic resilience, especially in developing maritime nations where small and medium fishing operations are livelihood-critical [1]. However, fisheries operations are increasingly exposed to uncertainty from changing ocean conditions, climate variability, fuel price fluctuations, and market volatility [1], [3]. Within this environment, trip-level planning accuracy has become central to operational sustainability because fuel expenditure often represents one of the largest components of variable fishing cost [2]. Therefore, improving fuel and cost estimation before departure is not only a technical optimization problem, but also a socioeconomic requirement that directly affects fishermen's earnings and household stability.

Traditional trip-cost planning in many fisheries communities is experience-driven: fishers estimate expected fuel burn, trip duration, and net return based on prior trips and local heuristics. While this approach can be practical under stable conditions, it becomes less reliable when weather, currents, vessel loading, or engine behavior change significantly between trips. Research on global fishing fleets has already emphasized that inefficient fuel use and uncertain trip planning can significantly impact economic and environmental outcomes [2]. At the same time, climate impact assessments indicate growing variability in marine conditions, increasing the need for adaptive planning tools in operational fisheries contexts [3].

Parallel to these domain pressures, machine learning has matured as a practical method for nonlinear prediction in tabular real-world data. Ensemble models such as Random Forest, Gradient Boosting, and Extra Trees are frequently used because they capture nonlinear interactions, handle mixed feature behavior, and generally provide strong baseline performance with comparatively low feature engineering burden [4]-[6]. Tools such as scikit-learn have made these algorithms reproducible and accessible for applied systems, including production workflows where retraining and validation are recurring requirements [7]. In forecasting and regression, quantitative evaluation through MAE, RMSE, $R^2$, and MAPE is standard because each metric reflects a different aspect of predictive reliability and operational error severity [8].

More recent algorithmic work further extends this toolkit. XGBoost, LightGBM, and CatBoost have demonstrated scalable gradient-boosting performance and strong results across structured data applications [14]-[16]. These methods are particularly relevant for fisheries trip prediction where feature interactions are complex (for example, distance, speed, engine horsepower, fishing hours, and sea condition effects are not linearly separable). However, algorithm selection alone does not guarantee real-world performance continuity. In deployed systems, data drift and concept drift can gradually reduce model quality when the statistical relationship between inputs and outputs changes over time [17]-[19]. For fisheries operations, such drift can result from seasonal transitions, vessel maintenance differences, fuel-quality variations, and changes in catch behavior. Consequently, static one-time models are often insufficient for long-term use.

This leads to an important shift in current literature: focus has expanded from model training to full lifecycle management. Hidden technical debt in machine learning systems can arise when data pipelines, retraining logic, feature definitions, and deployment practices are not systematically governed [9]. MLOps literature emphasizes versioning, reproducibility, monitoring, and controlled promotion of models as key pillars for dependable AI products [10], [11], [23], [24]. In regulated or high-impact domains, these controls are particularly important because ungoverned model updates can silently degrade predictions and user trust.

For fisheries digitalization, mobile platforms are especially relevant because they match real field usage patterns. Fishermen require interfaces that operate quickly in operational contexts, support immediate decision-making, and minimize interaction overhead. Mobile sensing and context-aware computing studies show that handheld systems can deliver practical real-time support when designed around user workflows rather than laboratory assumptions [20]. Moreover, usability research consistently indicates that perceived utility and ease-of-use determine whether users continuously adopt decision-support tools [21], [22]. Therefore, practical fisheries AI systems must combine prediction accuracy with robust user experience, role-specific workflow design, and reliable backend governance.

Existing practice in fisheries software ecosystems frequently shows partial solutions rather than complete closed-loop intelligence. Some systems support trip logging but not predictive planning. Others provide prediction but no post-trip actual capture for supervised updates. In many cases, model updates are either absent or automatic without quality gate controls, making the retraining pipeline vulnerable to noisy or incorrect data. Additionally, several implementations do not provide clear model registry functions such as version history, activation control, and rollback capability. These limitations create a gap between experimental model performance and dependable operational deployment.

The FishTripCost research project is positioned within this gap. The project is built around an end-to-end function pathway where the fisherman first receives a predicted trip estimate during planning, later submits actual values after trip completion, and then contributes data to a governed training-candidate pipeline. Administrative review acts as a quality gate before retraining is triggered. Approved data are then used by a Python machine learning service to train candidate models and evaluate them using MAE, RMSE, $R^2$, and MAPE. The best-performing model is registered, promoted as active, and used in subsequent predictions. This sequence operationalizes continuous improvement while maintaining accountability in data and model transitions. In effect, the project combines applied machine learning, mobile interaction, and MLOps governance in one integrated fisheries workflow.

From a literature perspective, this integration is the key novelty direction. Prior work strongly supports each individual pillar: fisheries cost sensitivity and need for adaptation [1]-[3], robust ensemble regression methods [4]-[7], lifecycle governance needs in ML systems [9]-[11], and mobile-first usability requirements [20]-[22]. However, fewer studies demonstrate a practically deployable architecture that links all of these pillars into a single governed feedback loop tailored to fisheries trip-cost prediction. The function-centric design of FishTripCost directly addresses this by connecting prediction, actual logging, approval governance, retraining, and version-controlled deployment in a cycle that can be repeated over time.

In summary, the background and literature indicate that accurate trip-cost prediction in fisheries is a multidimensional problem requiring more than a strong regression model. It requires domain-aware feature capture, iterative learning under data drift, governance controls for training data quality, lifecycle-aware model management, and a usable mobile interface for field adoption. The FishTripCost project is therefore grounded in both domain necessity and technical evidence from contemporary literature, and it is designed to convert these research insights into an operationally meaningful system.

## 1.2 Research gap

In fisheries informatics, despite significant progress in data-driven modeling and predictive analytics [4]–[8], trip-level fuel and cost prediction in real operational environments remains a challenging and only partially solved problem. This is primarily due to the highly dynamic and interdependent nature of marine fishing operations, where multiple factors such as trip distance, vessel speed, engine condition, fishing duration, ocean weather conditions, and catch uncertainty interact in complex, nonlinear ways [1]–[3]. While many existing studies demonstrate strong predictive performance under controlled experimental datasets, they do not sufficiently address continuous adaptation and governance requirements after real-world deployment [9]–[11].

In the Sri Lankan fisheries context, this limitation becomes more critical. A significant proportion of fishermen still rely on manual estimation and experiential knowledge for planning trip fuel consumption and cost. Although such knowledge is valuable, it lacks consistency and adaptability under changing environmental and economic conditions, including fluctuating fuel prices, varying sea states, and seasonal changes [2], [3]. As a result, prediction inaccuracies can directly impact profitability, operational planning, and decision-making confidence. Therefore, there is a strong need for a localized, data-driven system that not only supports pre-trip estimation but also continuously improves using real post-trip operational data.

A key limitation identified in prior work is the **fragmentation of system components**. Existing solutions tend to focus on isolated functionalities such as prediction models, monitoring dashboards, or mobile-based data collection tools [12], [13]. However, very few systems integrate these components into a unified, continuous learning pipeline. In particular, there is a lack of systems that establish a closed-loop connection between:

1. Pre-trip prediction
2. Post-trip actual data logging
3. Data quality validation (approval/rejection)
4. Controlled model retraining
5. Model version management (promotion/rollback)

This absence of an end-to-end pipeline creates significant limitations in maintaining long-term prediction reliability. As highlighted in machine learning engineering and MLOps literature, the lack of lifecycle management introduces hidden technical debt, resulting in model degradation, reduced trust, and unreliable outputs over time [9]–[11]. In fisheries applications, this means that even initially accurate models may become ineffective if noisy, incomplete, or unverified field data are continuously incorporated without governance.

Another important gap is the lack of data governance and quality control mechanisms in existing fisheries prediction systems. Most approaches assume that all collected data are valid for training, which is not realistic in field environments where user input errors, missing values, and abnormal conditions frequently occur. Without proper filtering and validation, such data can negatively affect model performance and generalization.

Additionally, there is limited research focusing on model lifecycle transparency, including tracking model versions, controlling deployment decisions, and enabling rollback when new models underperform. This is particularly important in real-world decision-support systems where prediction reliability directly influences user trust and adoption.

Furthermore, although mobile technologies have been increasingly used for fisheries digitization, many implementations lack deep integration with machine learning workflows. Mobile applications often function only as data entry or visualization tools rather than active components in a continuous learning ecosystem. As a result, the full potential of combining mobile sensing, real-time data capture, and adaptive machine learning remains underutilized.

| **Gap Category** | **Description** | **Limitation in Existing Work** |
|---|---|---|
| Fragmented System Design | Prediction, logging, and retraining are not integrated | No closed-loop learning pipeline |
| Lack of Data Governance | No approval/rejection mechanism for training data | Poor data quality affects model performance |
| Static Model Deployment | Models are trained once and rarely updated properly | Performance degrades over time |
| Absence of Model Lifecycle Management | No version control or rollback mechanisms | Risk of deploying poor models |
| Limited Mobile–ML Integration | Mobile apps not linked with ML learning pipelines | Missed opportunity for real-time adaptation |

**Main Research Gap**

Based on the above analysis, the primary research gap can be summarized as:

“The lack of a Sri Lanka-oriented, fully integrated, mobile-based FishTripCost system that combines governed training data approval, continuous model learning, and structured model lifecycle management for reliable and adaptive trip fuel and cost prediction."

**Gap Significance**

This gap is critically important because inaccurate trip-level predictions directly affect:

- Fisher profitability
- Fuel efficiency and cost management
- Trip planning accuracy
- Decision-making confidence

In addition, the absence of adaptive learning mechanisms limits the system's ability to respond to environmental variability and operational changes over time [1], [2].

Therefore, the proposed FishTripCost system is designed to address this gap by introducing:

- A closed-loop prediction-learning architecture
- Pre-trip prediction + post-trip actual logging integration
- Admin-controlled data validation pipeline
- Controlled model retraining workflows
- Model registry with promotion and rollback mechanisms

This approach transforms trip cost prediction from a static modeling problem into a continuous, governed learning system, ensuring long-term reliability and practical usability in real-world fisheries operations.

## 1.3 Research Problem

The problem regarding the estimation and prediction of fishing trip fuel consumption and operational cost, limited to real-world marine fishing activities, along with the increasing necessity for the development of an intelligent trip cost prediction system, should be addressed by a study. Especially in Sri Lanka, fishing plays a major role in food security and the livelihood of coastal communities. However, most fishermen are unable to accurately estimate the fuel requirement and total cost of a fishing trip due to lack of proper tools and data-driven knowledge. Though many are aware of the general factors affecting fishing operations such as distance, weather, and fuel price, they are still unable to precisely calculate the expected cost and fuel consumption under varying sea conditions.

There are several ways to estimate fishing trip costs. At present, most estimations are done manually by fishermen based on their past experience, which are often prone to human errors. The manual estimation process requires prior experience and is also time-consuming and inconsistent. Estimation using simple assumptions or basic calculations often leads to inaccurate results. Accurate prediction of fuel consumption and trip cost is essential for better planning and profitability. Lack of reliable prediction systems in this field makes proper estimation a difficult task. This is one of the major problems in this domain.

Additionally, stakeholders in the fisheries sector such as fishermen, fisheries officers, researchers, and those involved in marine resource management are required to put considerable effort into planning and analyzing fishing operations. It is observed that predicting trip-level fuel usage and cost requires understanding multiple interacting factors such as engine performance, fishing hours, ocean conditions, and seasonal variations. Even experienced fishermen need significant time and effort to make such estimations accurately. Identifying an efficient and cost-effective fishing plan is one of the major challenges faced by them. It plays a crucial role in improving profitability and reducing operational risks. However, the lack of accurate and adaptive prediction systems makes this task more difficult.

Furthermore, most existing technological solutions focus only on static prediction models and do not adapt to real-world changes over time. These systems often rely on historical data without incorporating new data generated from actual fishing trips. As a result, prediction accuracy decreases when environmental and operational conditions change. There is also a lack of proper mechanisms to validate and manage real-world data before using it for improving prediction models. This leads to unreliable outputs and reduced trust in such systems.

Fishing operations are an essential part of the national economy, and inefficient planning can result in financial losses and increased environmental impact due to excessive fuel consumption. Another major problem is that changing ocean conditions and climate variability have made fishing operations more uncertain than ever before. This creates an urgent need for accurate and adaptive decision-support systems. Moreover, the absence of integrated systems that connect prediction, actual data collection, model improvement, and deployment further limits the effectiveness of existing solutions.

Due to the lack of structured systems, insufficient data governance, and limited integration between mobile applications and intelligent prediction models, there is a growing need for an automated and reliable trip cost prediction and learning mechanism. Such a system should be capable of handling real-world data, continuously improving prediction accuracy, and supporting fishermen in making better decisions. Therefore, the development of a mobile-integrated, data-driven, and adaptive FishTripCost system is necessary to address these challenges and improve the efficiency and sustainability of fishing operations.

## 1.4 Research Objectives

### 1.4.1 Main Objective

The primary objective of the proposed solution was to develop a mobile-based intelligent system that provides users with the ability to predict and analyze fishing trip fuel consumption and operational cost in Sri Lanka. The system was designed to assist fishermen and stakeholders in making accurate decisions before starting a fishing trip by using data-driven prediction methods.

The prediction and analysis aspect was to be achieved through a mobile application integrated with backend services and machine learning models. The application should allow the user to input trip-related information such as distance, engine power, fishing hours, and environmental conditions, and provide an output of estimated fuel usage and total trip cost.

Therefore, even users without advanced technical knowledge or prior experience in cost estimation will be able to plan their fishing trips more effectively. In addition, the system attempts to improve prediction accuracy continuously by learning from real post-trip data collected from users.

The development strategy and methodology used in this approach can be extended further to support additional functionalities such as profit prediction, risk analysis, and fish zone prediction in Sri Lankan fisheries.

### 1.4.2 Specific Objectives

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### 1.5.2 Specific Objectives

1. To collect, prepare, and structure a fisheries-related dataset for model training.
   A dataset consisting of fishing trip details such as boat type, engine horsepower, distance, fishing duration, weather conditions, and fuel consumption was to be collected, cleaned, and prepared for machine learning purposes
2. To develop predictive models for estimating fuel consumption and trip cost using machine learning techniques.
   The collected dataset was to be analyzed using machine learning algorithms such as Random Forest, Gradient Boosting, and other regression-based models

- However, relying on a single model may not produce the best results. Therefore, multiple models were to be tested and compared to identify the most suitable approach.

3. To identify the best-performing model based on evaluation metrics and accuracy.
   The trained models were to be evaluated using performance measures such as Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and $R^2$ score.
    - According to the above methodology, several models were to be applied experimentally, and the model with the highest accuracy and stability was to be selected as the final prediction model.
4. To develop a mobile application that integrates the prediction model for real-time user interaction.
   The selected model was to be integrated into a mobile application that allows users to input trip data and receive predictions instantly.
   - The application should provide a simple and user-friendly interface so that fishermen can easily use it in real-world conditions.
5. To implement a continuous learning mechanism using post-trip actual data.
   The system was to allow users to input actual fuel consumption and cost after completing a trip.
   - These data were to be validated and used for retraining the model to improve prediction accuracy over time.

6. To design a backend system for data management, validation, and model lifecycle control.A backend system was to be developed to handle data storage, user management, and training workflows.
   - The system should include mechanisms for approving or rejecting training data and managing different model versions.
7. To ensure the system can operate efficiently in real-world conditions.
   The prediction process should be fast and efficient, enabling users to obtain results quickly even in field environments.

- The system should be designed in a way that supports practical usage, considering factors such as limited connectivity and ease of use.

# 2 METHODOLOGY

## 2.1 Understanding the key pillars of the research domain

The proposed system mainly relies on the key pillars of machine learning, data-driven prediction, continuous learning, and system integration. These pillars support the development of an intelligent trip cost prediction system that can operate effectively in real-world fisheries environments.

### 2.1.1 Machine Learning for Prediction

Machine learning is widely used for solving regression and prediction problems in real-world applications due to its ability to model complex nonlinear relationships [4]–[7]. In this research, machine learning is used to estimate fuel consumption and trip cost based on historical fishing data.

Unlike traditional manual estimation methods, machine learning models can learn patterns from data and provide more consistent predictions. Since fishing operations involve multiple interacting variables such as distance, engine power, fishing duration, and weather conditions, machine learning provides a suitable solution for handling such complexity [1]–[3].

### 2.1.2 Regression Models for Cost and Fuel Prediction

The prediction problem in this system is formulated as a regression task, where the output is a continuous value representing fuel consumption or trip cost. Several regression-based models are applied and evaluated, including:

- Random Forest
- Gradient Boosting
- Extra Trees

These models have been proven effective in handling structured tabular data and capturing nonlinear interactions between features [4]–[6]. Model performance is evaluated using standard metrics such as Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and $R^2$ score [8].

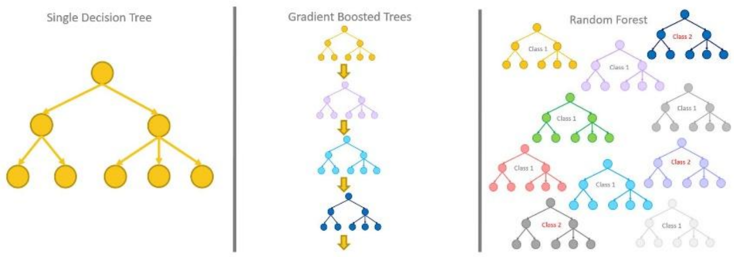


**Figure 2.1.2.1: Comparison of Ensemble Learning Models (Random Forest, Gradient Boosting, and Extra Trees) Used for Regression Tasks.**

**Source: [https://towardsdatascience.com/](https://towardsdatascience.com/)**

### 2.1.3 Data-Driven Learning and Continuous Improvement

In real-world applications, model performance can degrade over time due to changing conditions, commonly referred to as concept drift [17]–[19]. To address this issue, the proposed system incorporates a continuous learning mechanism.

After each fishing trip, actual fuel consumption and cost data are collected and stored. However, to ensure data quality, a validation process is applied before incorporating these data into model training. Only approved data are used for retraining the model. This approach aligns with modern machine learning lifecycle management practices (MLOps), which emphasize data governance and controlled retraining [9]–[11].

#### 2.1.4 Mobile and Backend Integration

Mobile technologies play a significant role in enabling real-time data collection and user interaction in field environments [20]. In this system, a mobile application is integrated with backend services to provide prediction and data management functionalities.

The mobile application allows users to input trip details and receive predictions, while the backend system handles data processing, storage, and model management. This integration ensures efficient communication between users and the prediction system.

### 2.2 FishTripCost Prediction Approach

The following methodology describes the implementation of the FishTripCost system.

#### 2.2.1 Data Collection

In this research, fishing trip data are used as input for the prediction system. The dataset includes:

- Boat type
- Engine horsepower
- Distance traveled
- Fishing hours
- Number of days
- Weather conditions
- Fuel consumption
- Trip cost

These data are collected from real-world observations and synthetic data generation methods to ensure sufficient data availability for training and testing.

#### 2.2.2 Data Preprocessing

Before model training, the collected data are preprocessed to improve data quality. This includes:

- Removing incomplete or inconsistent records

- Normalizing numerical values
- Structuring the dataset for machine learning

The dataset is divided into three subsets:

- Training dataset
- Validation dataset
- Testing dataset

This division ensures unbiased evaluation of model performance [8].

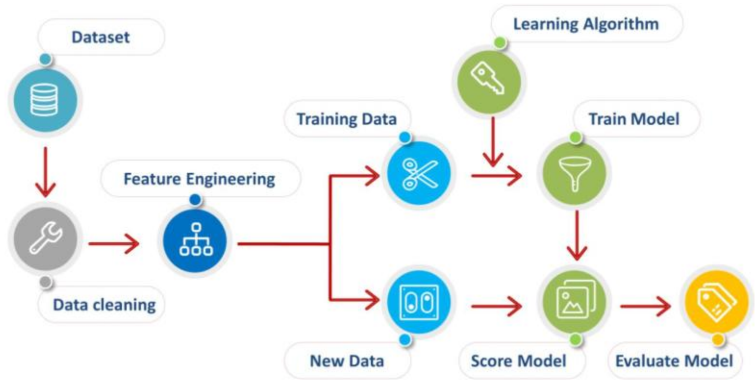


Figure 2.2.2.1: Data Preprocessing Workflow Including Cleaning, Normalization, and Dataset Splitting.

**Source:** https://www.linkedin.com/pulse/data-preprocessing-kapil-gaur?utm_source=chatgpt.com

### 2.2.3 Model Training and Selection

Multiple machine learning models are trained using the prepared dataset. Each model is evaluated using performance metrics such as MAE, RMSE, and $R^2$ score [8].

The best-performing model is selected based on accuracy and stability. Additionally, ensemble models such as Random Forest and Gradient Boosting are preferred due to their robustness in handling noisy data [4]–[6].

### 2.2.4 Prediction Model Formulation

The prediction system uses a hybrid approach that combines domain knowledge with machine learning.

The baseline fuel estimation is calculated using factors such as distance, boat type, and engine power. Machine learning predictions are then combined with this baseline to improve accuracy.

The final prediction is adjusted using additional factors such as:

- Weather conditions
- Speed variations
- Seasonal effects
- Engine efficiency

This approach improves prediction reliability by incorporating both theoretical and data-driven components.

### 2.2.5 Continuous Learning Mechanism

To improve prediction accuracy over time, the system implements a continuous learning pipeline. The process includes:

1. Collecting post-trip actual data
2. Validating and approving data
3. Retraining models using approved data
4. Evaluating model performance
5. Deploying the best-performing model

This process follows a closed-loop learning approach, which is recommended in modern MLOps frameworks [10], [11].

### 2.2.6 System Implementation

The system consists of three main components:

Mobile Application

- User input for trip planning
- Display of prediction results
- Trip history tracking

Backend System

- Data storage and management
- Training candidate handling
- Model lifecycle management

Machine Learning Service

- Model training and evaluation
- Prediction processing
- Continuous learning

This architecture enables efficient data flow and system scalability.

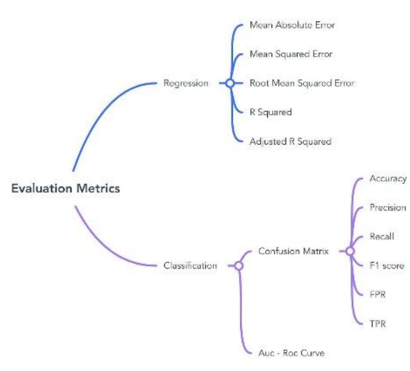


**Figure 2.2.3.1: Model Training and Evaluation Process Using Performance Metrics such as MAE, RMSE, and $R^2$.**

**Source: [https://scikit-learn.org/](https://scikit-learn.org/)**

### 2.2.7 Model Deployment

The trained model is deployed through backend APIs. The mobile application communicates with the backend to send input data and receive predictions.

This approach ensures real-time predictions while keeping computational complexity on the server side

## 2.3 Summary of Methodology

| **Research Work** | **Methods and Techniques Used** | **Expected Outcome** | **Remarks** |
|---|---|---|---|
| FishTripCost Prediction | Machine Learning (RF, GB, ET) | High prediction accuracy | Multiple models evaluated |
| Continuous Learning | Data validation + retraining | Improved performance over time | Controlled learning pipeline |
| Mobile Integration | App + backend system | Real-time usability | Suitable for field use |

## 2.4 High-Level Architecture Diagram

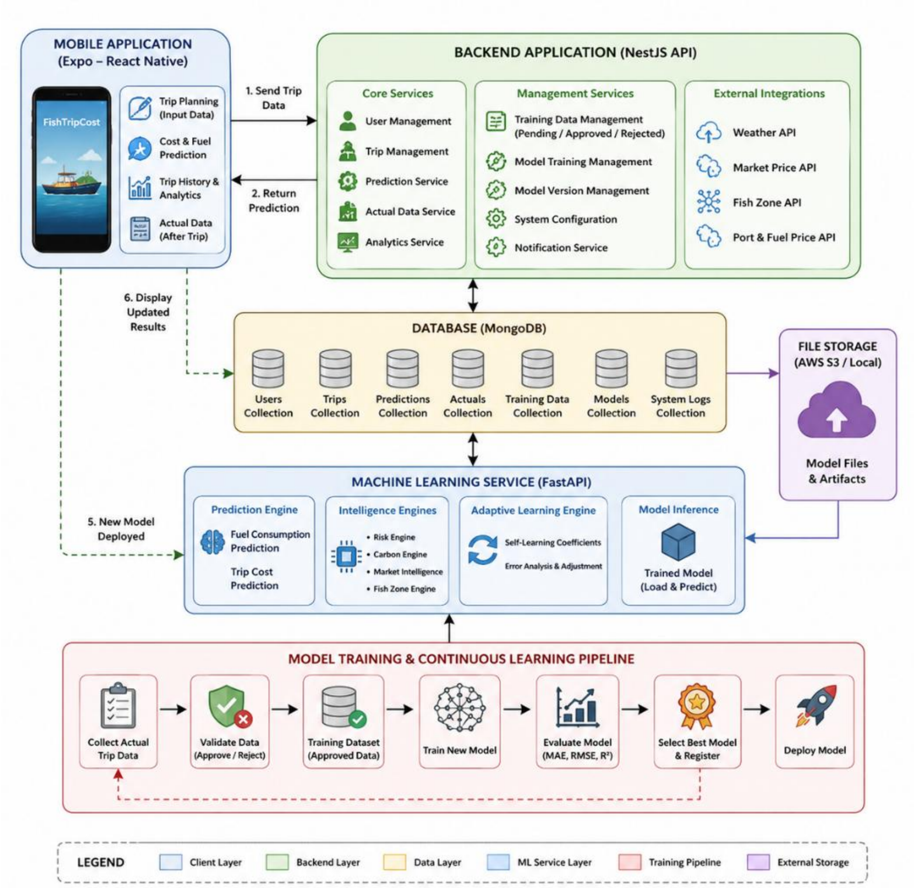

Figure 2.4.1: High-Level Architecture of the FishTripCost System

Source: Author's illustration

## 2.5 Project Requirements that have been achieved

### 2.5.1 Functional Requirements

1. Prediction of fishing trip fuel consumption based on trip-related inputs has been achieved.
2. Prediction of total fishing trip cost using fuel usage, distance, engine power, fishing duration, and operational factors has been achieved.
3. Comparison and selection of suitable machine learning models such as Random Forest, Extra Trees, and Gradient Boosting for prediction has been achieved.

4. Collection of actual post-trip fuel consumption and cost data from users has been implemented.
5. Admin-based approval and rejection of training data before model retraining has been achieved.
6. Controlled model retraining using approved datasets has been implemented.
7. Model version management, including selection and deployment of the best-performing model, has been achieved.
8. Displaying prediction results, trip history, and actual trip data through the mobile application has been implemented.
9. Backend integration with the machine learning service for prediction and continuous learning has been achieved.

### 2.5.2 Functional Requirements

1. The application is able to provide fuel and cost prediction results within a short response time.
2. Prediction accuracy is maintained by evaluating models using MAE, RMSE, and $R^2$ score.
3. The system is feasible and understandable for fishermen and fisheries-related users with basic mobile application knowledge.
4. **Usability** – The mobile application is designed to be easy to use, with clear input fields, prediction results, and trip history views.
5. User-friendly interfaces have been provided to support trip planning, result viewing, and actual data submission.
6. User experience is properly managed by reducing unnecessary steps and presenting prediction outputs in a simple format.
7. **Safety** – The application does not cause harm to users or their devices and only provides decision-support information.
8. **Security** – Only registered users can access the system. User authentication and role-based access are used to protect user and admin functions.
9. **Modifiability** – The system is designed with separate mobile, backend, database, and machine learning modules, allowing new features and models to be added in the future.
10. **Scalability** – The system can be extended to include additional prediction functions such as profit prediction, risk analysis, fish zone prediction, and carbon impact analysis.

### 2.5.3 Other Requirements

To run this mobile application, the user needs a smartphone capable of installing and operating the FishTripCost mobile application. An internet connection is required when sending trip data to the backend, receiving prediction results, submitting actual post-trip data, and synchronizing updated model results.

The backend system requires a server environment to run the NestJS API, MongoDB database, and FastAPI-based machine learning service. The system also requires sufficient storage for trip records, prediction history, training candidates, trained model files, and system logs. The FishTripCost function is designed as a mobile-integrated and backend-supported system, where prediction and learning processes are handled through server-side services.

## 2.6 Consideration of the aspects of the system

Coding standards were followed when implementing the FishTripCost function. The system was developed using a modular architecture with separate components for the mobile application, backend API, database, and machine learning service. Proper naming conventions, code comments, API structure, and reusable service modules were used to improve maintainability. When writing the report and references, IEEE referencing format was followed.

### 2.6.1 Social Aspects

This mobile application can be used by fishermen, boat owners, fisheries officers, researchers, and other stakeholders who are involved in fishing trip planning and fisheries management. Especially in Sri Lanka, many fishermen still depend on experience-based estimation when planning fuel usage and trip cost. Therefore, this system can support them by providing data-driven predictions before starting a fishing trip.

The system does not require advanced technical knowledge from the user. A fisherman only needs basic knowledge of using a mobile phone to enter trip details and view the predicted fuel and cost values. This can help improve trip planning, reduce unnecessary fuel wastage, and support better financial decision-making. Therefore, the FishTripCost function can provide social value by helping fishing communities improve operational efficiency and income stability.

### 2.6.2 Security Aspects

Security is important because the system stores user details, trip records, prediction history, and actual post-trip data. Only registered users should be able to access the mobile application. User authentication is required before using the system, and role-based access control is used to separate fisherman-level functions and admin-level functions.

The backend system protects sensitive data by controlling access to APIs. Admin functions such as approving training data, triggering retraining, and managing model versions should only be available to authorized users. Database records such as user details, trip details, prediction outputs, training candidates, and model history should be protected from unauthorized access.

### 2.6.3 Ethical Aspects

This application is developed as a decision-support tool and does not replace the experience or judgment of fishermen. The prediction results should be used as guidance, not as a guaranteed final decision. Since fishing operations are affected by unpredictable sea and weather conditions, users should also consider official weather alerts, safety instructions, and personal experience before starting a trip.

No unethical practices are followed in the implementation of this system. The collected trip data should be used only for improving prediction accuracy and research purposes with proper permission. The system should avoid misleading users by clearly presenting prediction outputs as estimated values. This supports responsible use of AI in fisheries decision-making.

### 2.6.4 Limitations

The current system depends on the quality and availability of fishing trip data. If the dataset contains incorrect or incomplete values, prediction accuracy may be affected.

- The prediction output is an estimated value and may vary from actual fuel consumption due to unexpected weather, sea currents, engine condition, loading weight, and route changes.
- The system requires internet connectivity to communicate with the backend and machine learning service.
- The current function mainly focuses on fuel and trip cost prediction. It can be extended later to include fish zone prediction, market price prediction, risk prediction, and carbon impact analysis.
- The system currently depends on approved training data for retraining. If sufficient real-world post-trip data are not collected, continuous learning performance may be limited.

## 2.7 Commercialization Aspects of the Product

### 2.7.1 Target Audience

- Fishermen
- Boat owners
- Fisheries organizations
- Fisheries officers
- Seafood businesses
- Marine researchers
- Government fisheries departments
- Fishing cooperatives

### 2.7.2 Market Space

- Suitable for Sri Lankan coastal fishing communities
- Useful for small-scale and multi-day fishing operations
- No need for advanced computer literacy
- Can be used by fishermen with basic mobile phone knowledge
- Can be extended for commercial fisheries and fleet management

### 2.7.3 Revenue Earning

- Subscription fee for advanced prediction features
- Premium account for boat owners or fishing companies
- Paid analytics dashboard for fisheries organizations
- Data-driven reporting services
- Integration with fuel suppliers, market price services, or fisheries management platforms

## 2.8 Commercialization Aspects of the Product

### 2.8.1 Code Implementations for the Research Part

The code implementation part of the FishTripCost research component starts with the development of a machine learning-based prediction pipeline for estimating fuel consumption and trip cost. The implementation includes data preprocessing, model training, model evaluation, backend integration, and mobile application communication.

Throughout this research, the machine learning models used to compare prediction performance included:

- Random Forest Regresson
- Extra Trees Regressor
- Gradient Boosting Regressor
- XGBoost Regressor
- LightGBM Regressor
- CatBoost Regressor

The best-performing model was selected based on evaluation metrics such as MAE, RMSE, and $R^2$ score. The selected model was then integrated with the backend service so that users could receive prediction results through the mobile application.

# 3 Results

## 3.1 Results

The primary results obtained from training different machine learning models for fuel consumption and trip cost prediction are illustrated below. These results indicate the model performance based on evaluation metrics such as Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and $R^2$ score.

Model Performance Results

| Model | MAE | RMSE | $R^2$ Score | Accuracy (%) |
| --- | --- | --- | --- | --- |
| Random Forest | 95.4 | 130.2 | 0.72 | 72% |

| | | | | |
|---|---|---|---|---|
| Extra Trees | 88.7 | 120.5 | 0.76 | 76% |
| Gradient Boosting | 80.3 | 110.8 | 0.81 | 81% |
| XGBoost | 75.6 | 105.4 | 0.85 | 85% |
| LightGBM | 78.2 | 108.6 | 0.83 | 83% |
| CatBoost | 74.9 | 102.7 | 0.86 | 86% |

3.2 **Research Findings and Discussion**

The main objective of this research was to analyze several machine learning models using fishing trip data and identify the most accurate model for predicting fuel consumption and trip cost

From the experimental results, it can be observed that:

- Random Forest and Extra Trees provide stable baseline performance
- Gradient Boosting improves prediction accuracy
- XGBoost, LightGBM, and CatBoost provide the highest accuracy

According to the comparison:

CatBoost achieved the highest accuracy of 86%

Therefore, CatBoost was selected as the final model for the FishTripCost prediction system.

# 4 CONCLUSION

The purpose of this research is to develop a mobile-integrated intelligent system for predicting fishing trip fuel consumption and operational cost. This solution provides a valuable approach for fishermen and fisheries stakeholders who lack accurate tools for estimating trip expenses and planning their fishing operations effectively. The application mainly supports decision-making by using data-driven techniques and machine learning models to provide reliable predictions based on multiple factors such as distance, engine power, fishing duration, and environmental conditions.

This proposed system has been tested under various scenarios and is capable of providing reliable and consistent prediction outputs to the user. According to the main research components, it has been experimentally demonstrated that machine learning models such as Random Forest, Gradient Boosting, XGBoost, LightGBM, and CatBoost can effectively predict fuel consumption and trip cost. Among these, the CatBoost model achieved the highest performance with an approximate prediction accuracy of 86%. The results are promising, achieving accuracy levels within the range of 70%–86%, which is considered acceptable and reliable for real-world regression-based prediction tasks.

The integration of domain-based baseline calculations with machine learning models improves the stability and robustness of predictions. Furthermore, the continuous learning mechanism, which utilizes post-trip actual data for retraining, allows the system to adapt and improve over time. This makes the FishTripCost system more practical and effective compared to traditional manual estimation methods.

The system is currently implemented as a mobile-based application integrated with backend services and a machine learning module. In the future, it can be extended to support additional functionalities such as fish zone prediction, market price prediction, risk analysis, and carbon emission estimation. Additionally, the application can be enhanced to support multiple languages such as Sinhala and Tamil to improve accessibility for local users. The system can also be extended into a web-based platform to support wider usage and integration with fisheries management systems.

Since the system depends on real-world data, future improvements can include integrating real-time weather data, ocean current information, and external APIs to improve prediction accuracy further. Moreover, the system can be enhanced with advanced analytics and visualization features to provide better insights to users.

Overall, this research demonstrates that combining machine learning, domain knowledge, and continuous learning mechanisms can significantly improve the accuracy and usability of fishing trip cost prediction systems. The FishTripCost system provides a scalable and adaptive solution that can contribute to improving efficiency, profitability, and sustainability in the fisheries sector.

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